

Semantic business process space for intelligent management of sales order business processes

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Abstract A company's competitiveness relies heavily on its business processes and accurate knowledge to execute its business processes with agility and efficiency. Business Process Management (BPM) initially promised to provide the business world with suitable tools and techniques for successful BPM without help from the IT world. However, the current practice of BPM has several fundamental problems, including difficulty with automatic discovery and the integration of business processes across organizations. Understanding that the main cause of these problems lies in the lack of semantics on business process, we first define a variety of business process ontologies in order to build a *semantic business process space (SBPS)* for the limited area of *sales order*. We then explain how the SBPS satisfies the requirements for successful implementation of semantic BPM (SBPM) and demonstrate with a scenario how SBPM can be realized in the environment of SBPS. Our novel approach will reduce the time and cost necessary for the development of a new business process in a fast-changing environment and provide practitioners with useful insights into the proper implementation of the SBPM. Although our paper defines semantic business process knowledge for only a limited domain, its insights can be readily extended to other areas of business.

Keywords Semantic business process management · Semantic business process space · Business process management · Semantic web services · Ontology

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1 Introduction

Companies that want to improve their performance and survive ever-increasing competition can benefit from analyzing the business processes of a leading company in the same industry in order to catch up with it or to radically redesign their business processes from scratch (Davenport and Short 1990; Hammer 1990). Benchmarking and Business Process Reengineering (BPR), popular in the 1990s, were essentially taken over in the early 2000s by Business Process Management (BPM), a leading technology for pursuing the agility and efficiency required for competitiveness (Dumas et al. 2005; Smith and Finger 2003; van der Aalst et al. 2003).

Initially, BPM promised to provide the business world with suitable tools and techniques with which to design, modify, execute, and analyze business processes without help any from the IT world (Smith and Finger 2003). However, current techniques for BPM still suffer from fundamental problems that must be resolved before BPM can fulfill its promise. These problems include difficulty in integrating business processes across organizations, difficulty in querying and reusing all the facts associated with business processes, non-automatic transformation from a business process model to an executable workflow model, irrelevant web services resulting from the lack of semantic specification in WSDL, and the lack of semantic description for dynamic discovery and automatic composition of Web services in BPEL specification.

Having recognized that these problems occur because of the lack of semantic information, pioneering researchers in the field have suggested Semantic BPM (SBPM) as a solution (Hepp et al. 2005; Hepp and Roman 2007; Pedrinaci and Domingue 2007; Wetzstein et al. 2007). However, most of these researchers have described what is

required for the implementation of SBPM in non-technical terms or have focused on resolving partial problems occurring during the BPM lifecycle, neither of which is helpful to practitioners who want to implement SBPM comprehensively in technical terms. Thus, the objective of this paper is to provide practitioners with specific insights into the proper implementation of SBPM. The core of our approach to resolving the problems of current BPM practice is to define the various ontologies of Business Process Space (BPS¹) that is shared by both the business world and the IT world and to show that the *Semantic Business Process Space* (SBPS²) of such ontologies allows a number of tasks that have been processed manually *by humans* to be processed automatically or semi-automatically *by machine*.

SBPS contains two layers of ontologies: Generic Business Process Ontology (GBPO) and Semantic Business Process Model/Semantic Web service (SBPM/SWS). The GBPO consists of five sub-ontologies—business process ontology, organization ontology, resource ontology, domain knowledge ontology and service ontology—and the relationships among them. The SBPM/SWS consists of a variety of business processes (e.g., sales order process and delivery process), semantic web services, and mapping between the business processes and their corresponding services. The generic concepts and relationships captured in GBPO are inherited to the specific business processes in SBPM/SWS.

Among the several ontological engineering methodologies that are frequently referenced, including METHONTOLOGY (Fernández-López et al. 1997), Cyc (Lenat and Guha 1990), Uschold and King (Uschold and King 1995), TOVE (Grüniger and Fox 1995), and On-To-Knowledge (Staab et al. 2001), we selected METHONTOLOGY to define our ontologies because it supports comprehensive activities for detailed definitions of ontology. Prior to defining the ontologies to be included in the GBPO, we referred to the conceptual ontologies (Hepp and Roman 2007) and other generic ontologies such as Enterprise Ontology (Uschold and King 1995) and TOVE ontology (Grüniger and Fox 1995). Based on these ontologies, we defined our ontologies, which were then maintained and will be iteratively evolved in accordance with the ontology development lifecycle of the METHONTOLOGY.

With these ontologies defined in SBPS, we can perform several functions in pursuit of the agile and efficient management of business processes. First, we can select and *compose* existing business processes, *modify* them into

a new one, or *design* a new one from scratch, if necessary. Whenever a new business process is defined in SBPS, its corresponding semantic web service must also be defined. Second, we can *create* a new business process by querying, instantiating and composing the business processes in SBPS. If we are to *execute* a business process, its corresponding semantic web service must be invoked and executed. Third, by using mining techniques to analyze the business processes found in the log data of actual business process execution, we can validate the business processes and/or identify a set of business processes as good candidates for a new business process.

The remainder of the paper is organized as follows. Section 2 describes the key technologies and the terminologies associated with SBPM. Section 3 introduces a number of recent studies associated with SBPM. Section 4 retraces the initial promises of BPM, explains the major problems that have been recognized in the current practice of BPM, and defines some requirements to solve the problems. Section 5 presents the overall structure of SBPS and defines our ontologies, GBPO and SBPM/SWS, on the basis of the structure with a focus on building SBPS for the sales order process in particular. Section 6 explains how SBPS satisfies the requirements for successful implementation of SBPM by solving the major problems found in Section 4. In Section 7, we utilize a scenario to demonstrate how SBPM can actually be realized in an SBPS environment. Finally, Section 8 concludes with our summary, and some considerations regarding our approach, including contributions and limitations.

2 Key technologies and terminologies for SBPM

The state of current BPM technologies helps to explain the rationale for SBPM technologies. Figure 1 shows the key technologies for BPM and SBPM. Web service technology (Booth et al. 2004) provides a standard means of distributed computing that is independent of platforms and programming languages, and thus supports interoperability between two parties that share some business processes. In the

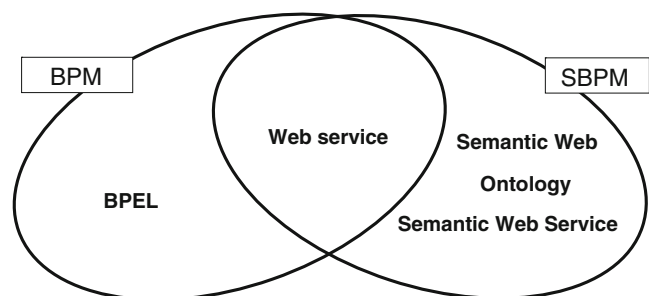


Fig. 1 Key technologies for BPM and SBPM

¹ Business Process Space is defined as a place where all the facts associated with intra—and inter-organization business processes are stored and maintained.

² SBPS is a semantic knowledge space in which all information associated with business processes is stored and maintained in machine-understandable form.

context of service-oriented architecture (SOA) (Erl 2005; Erl 2008), web service technology is one of the most powerful instruments for realizing efficient BPM (Leymann et al. 2002), because business processes can be executed after seamlessly combining proper web services (Liu et al. 2008).

Normally, a web service is specified in Web Service Description Language (WSDL³), which provides a means to design an interface through which a request can invoke a remote web service advertised by a provider with some information about the interface including its input and output parameters, operations, and service locations. In order to execute a business processes appropriately, it is first necessary to establish specifications for the composition and invocation of a set of coordinated web services in a workflow. BPEL⁴ specification supports describing an executable workflow model across multiple organizations by coding the definition of a *process* in XML format, including participating web services of *partners*, message exchange, and intermediate results of each *activity*. The resultant BPEL documents, together with the WSDL documents, are deployed into an execution engine in which completely automated tasks are executed by the web service and partially automated tasks are executed by human workers.

Currently, the two BPM technologies—web service and BPEL—play crucial roles in achieving the goal of BPM. However, both technologies have some problems in modeling web services, dynamic discovery of appropriate services, and automatic composition and execution of them to execute a business process. WSDL provides no means with which to describe semantics for input and output parameters in machine-understandable form. As a consequence, a number of irrelevant web services, as well as relevant ones, are discovered when a service requester searches an appropriate service among the available web services advertised by service providers (Cardoso and Sheth 2006; Studer et al. 2007). BPEL provides no semantics necessary for the communication among a set of coordinated web services in a workflow and allows only for a manual and static assignment of web services in design time, and thus BPEL cannot be utilized for dynamic discovery and automatic composition and execution of web services at run time (Mendling 2006).

To overcome these limitations of WSDL and BPEL, many researchers have conducted studies on semantic web services technology by incorporating semantics into web services (Bell et al. 2007; Cardoso 2006; Cardoso and Sheth 2006; Martin et al. 2004; McIlraith and Martin 2003; McIlraith et al. 2001; Preist 2004; Stollberg et al. 2006; Studer et al. 2007). Semantic web services technology is based on the semantic web technology (Berners-Lee et al. 2001) and the key concept underlying the idea of the semantic web is ontology, which is a specification that explicitly defines consensual knowledge in a specific domain in a formal way (Gruber 1993a, b; Horrocks 2008).

Typically, OWL (McGuinness and van Harmelen 2004) is used as the standard ontology language (McGuinness and van Harmelen 2004). OWL, which was derived from DAML+OIL⁵ (Hendler and McGuinness 2000) and developed by the W3C, has three different sublanguages—*OWL Full*, *OWL DL*, and *OWL Lite*—each of which has a different expressive power. *OWL Full* is the most expressive, *OWL Lite* is the least expressive and *OWL DL* falls between the two, but it is significantly more expressive than *OWL Lite*. *OWL DL* is based on *Description Logic* (Baader et al. 2002), which is a decidable fragment of *First Order Logic* and permits automated reasoning. Therefore, *OWL DL* allows us to check automatically how classes defined in ontology are subsumed and whether they are consistent. For these reasons, we selected *OWL DL* from the three sublanguages of OWL for the construction of our ontologies for BPS.

In OWL, a concept corresponds to a class, and a relationship corresponds to a property. A class defines a group of individuals with the same set of properties and axioms. A property is used to define a relationship between two individuals, called *ObjectProperty*, or between an individual and a literal value, called *DatatypeProperty*. Classes can be organized in a hierarchal structure, using the property *rdfs:subClassOf*, where the prefix *rdfs* indicates that the namespace of the property *subClassOf* is RDFS (Resource Description Framework Schema) (Brickley et al. 2004). RDFS provides a mechanism for specifying the structure of an RDF (Resource Description Framework) data model (Beckett and McBride 2004). RDF was developed to facilitate the automated processing of web resources in a machine-understandable form and to provide interoperability between applications. However, owing to the limited expressive power of RDFS, the extended ontology language, OWL, was developed.

³ WSDL is the standard specification of the World Wide Web Consortium (W3C) used to define a web service in standardized message exchange formats (Chinnici et al. 2007).

⁴ The initial version of BPEL, *BPEL4WS* (Business Process Execution Language for Web Services), was published in 2002, while the new one, *WS-BPEL* (Web Service Business Process Execution Language), was published in 2007 (<http://docs.oasis-open.org/wsbpel/2.0/wsbpel-v2.0.html>). The acronym “BPEL” stands for both versions.

⁵ DAML+OIL stands for DARPA Agent Markup Language and Ontology Inference Language. DARPA, in turn, stands for Defense Advanced Research Project Agency. DAML+OIL is the predecessor of OWL (<http://www.daml.org/language/>).

With ontology, semantic web services can be defined to add semantic information both to web services and to communication protocol.⁶ In other words, a semantic web service describes what a web service does and what sequence of messages is used to interact with other web services in machine-understandable form. Therefore, semantic web services enable machines to discover relevant web services and to compose and execute them automatically at runtime without human intervention (Sycara et al. 2003; Keller et al. 2005; Sivashanmugam et al. 2004; Charif and Sabouret 2006). Several languages—including Web Ontology Language for Services (OWL-S) (OWL-S Coalition 2008), Web Service Modeling Ontology (WSMO) (de Bruijn et al. 2005), WSDL-S (Akkiraju et al. 2005), and SAWSDL (Kopecký et al. 2007)—have been developed for the implementation of semantic web services.

Among them, OWL-S (formally, DAML-S⁷) supports a core set of constructs to define web services on the basis of OWL. OWL-S is composed of three interlinked sub ontologies, known as *service profile*, *service model*, and *service grounding*. In brief, a service profile is used to represent what the service does, for the purpose of advertising it with information on functional properties such as inputs, outputs, preconditions, and effects (IOPE) and on non-functional properties such as service name. A requester can locate a service and then construct a request to invoke the service using the profile. A service model defines how the service operates or how the service is interoperated with other services, using the concepts and relationships defined in one or a set of interrelated ontologies. A service grounding maps an atomic service to a WSDL operation and the I/O concepts of a service to the corresponding I/O message part of a WSDL operation, respectively. In this study, we selected OWL-S for the implementation of semantic web services, because a variety of tools⁸ are available that supports OWL-S.

3 Related work

The principal vision of SBPM is to support the implementation of agile and efficient BPM by querying SBPS for relevant business processes by combining BPM and

semantic web service technology (Hepp et al. 2005). A number of studies concerning academic issues associated with SBPM have been conducted in support of SBPM's ability to achieve this vision. Hepp and Roman (2007) proposed a representational architecture for the integration of dominant business modeling methodologies into the environment of SBPM, in addition to an upper-level ontology framework which includes a set of ontologies associated with business process (e.g., business process, organization and resources, business functions and logics, strategy) simply by listing informal competency questions in natural language that should be taken into account when building ontologies. Those competency questions suggested very helpful guidelines with which ontology engineers could implement ontologies associated with business processes for SBPM. Wetzstein et al. (2007) identified new functionalities that an SBPM system should provide on the conceptual level from the perspective of requirements, based on the BPM lifecycle (i.e., modeling, implementation, execution, and analysis phases). Many recent academic studies concerning SBPM have focused on resolving the problems that can arise in each phase of the BPM lifecycle.

Most of the research on issues related to the *modeling phase* has dealt primarily with the method for representing a semantically annotated business process model in machine-understandable form using ontology. In order to reduce typos and structural modeling errors, Betz et al. (2006) developed an auto-completion mechanism for a business process model that measures three similarities (i.e., syntactic similarity, linguistic similarity, and structural similarity) between the manual business process and the template business process. For the automatic synthesis of modeled business processes, Lautenbacher and Bauer (2006) developed an algorithm that computes three kinds of semantics (i.e., data semantics, functional semantics, and execution semantics) among the semantically annotated business processes. Thomas and Fellmann (2007) have developed a semantic extension of the Event-driven Process Chain (EPC) (Davis and Brabander 2007; Scheer et al. 2005) to remove the ambiguity inherent to the use of natural language in semi-formal business process models.

A great deal of research into various issues associated with the *implementation and execution phase* has focused on the translation of a business process model to an executable workflow model by discovering relevant web services dynamically and automatic composition and/or execution of them based on ontology. Cardoso and Sheth (2003) proposed a method for the efficient discovery of web services and the automatic composition of those web services by developing a set of algorithms that can increase the degree of precision of the discovery process, represent the operational metrics (e.g., quality of service (QoS)), and measure the degree of integration of the discovered web

⁶ Communication protocol consists of two types of constraints on message exchange: 1) *choreography* determines the constraints on the ordering of messages between two parties (a service requester and a service provider), and 2) *orchestration* specifies constraints on coordination among the interacting web services of organizations.

⁷ <http://www.daml.org/services/owl-s/>

⁸ The MINDSWAP Group at University of Maryland has been conducting a variety of studies and projects to develop tools and techniques such as OWL-S API and WSDL2OWLS which are useful for the implementation of semantic web services (<http://www.mindswap.org/>).

services and workflow. Guo et al. (2004) suggested a method for the conceptual mapping of a business process model in Fundamental Business Process Modeling Language (FBPML) (Chen-Burger et al. 2002) to a semantic web service in OWL-S in order to support the automatic discovery, composition, execution, and monitoring of web services. Haller and Oren (2006) developed an ontology, referred to as *m3pl*, for the conceptual mapping of internal workflow (e.g., IBM Websphere MQ Workflow) to choreography language (e.g., BPEL) in order to integrate different internal workflows between collaborative organizations⁹ by automatically generating uniform choreographies. In order to reuse the information of existing business processes and web services and to accelerate the development of an SOA application, Liao and Leung (2007) proposed a framework for modeling and developing SOA applications based on Business Process Ontology (BPO), which represented a simple conceptual hierarchy of business processes and the relationship between business processes and web services. However, they did not elucidate how the framework can be implemented based on the BPO.

Another stream of research regarding issues associated with the *analysis phase* of the BPM lifecycle has examined business processes using semantic web technology. Celino et al. (2007) introduced the technologies for semantic business process analysis, including process mining and reverse business engineering, and showed how the technologies can benefit from the use of semantic information in ontology. Pedrinaci and Domingue (2007) developed Event Ontology (EO) to support the monitoring of events at a specific point in time and Process Mining Ontology (PMO) to integrate the broad range of diverse knowledge that can be used to mine business processes for analyses of semantic business processes.

In contrast to the many academic studies, only a few attempts have been made to apply semantic web service technology to BPM from a *practical perspective* (Drumm et al. 2006; Preist et al. 2005). Preist et al. (2005) developed a demonstrator system to apply semantic web service technology to business processes in the domain of logistics and supply chain. Their system allowed a *service-request* agent to communicate with a *discovery-provider* agent to locate a relevant web service using ontology for semantic descriptions of web services offered by *service-provider* agents. In order to integrate business processes automatically among parties in the logistic domain, Drumm et al.

(2006) developed a mapping method that automatically generates message mapping by comparing the XML input and output schema of each party using a shared ontology. The generated mapping information, together with the semantic business process descriptions of each party, is used to integrate the business processes of all parties and to compose relevant web services that will execute the integrated processes.

Although quite a few academic studies concerning SBPM have been recently conducted, the focus of the majority of the research has been on resolving the partial problems arising in each stage of the BPM lifecycle, whereas the practical research efforts conducted in this field are still in their early stages. Therefore, more comprehensive research into resolving the major problems of BPM with semantic web technology should be undertaken.

Furthermore, in order to achieve agile and efficient BPM through direct querying and manipulating BPS from the managerial perspective, further research efforts are needed to describe, not just what is required for the implementation of SBPM, but also how SBPM can be implemented from a technical perspective. To that end, we attempt to achieve a comprehensive view of SBPM by defining the ontologies of SBPS to be shared by the business and IT worlds, and by demonstrating how to resolve major problems related to BPM using the ontologies. Table 1 contains a summary of related works.

4 BPM: Initial promises and current problems

4.1 Initial promises of BPM

BPM emerged in the early 2000s as a leading technology for the pursuit of high agility and efficiency in a dynamic environment. The aim of BPM is to manage the direct execution of business processes from a high-level perspective in order to reduce the cost and time spent in executing business processes (Smith and Finger 2003). To achieve this aim, BPM promised to offer stakeholders in the business world the tools and techniques to:

- provide a unified BPS against which queries could be issued to find relevant business processes
- reduce the time gap between the business world and the IT world
- implement and execute business processes directly without help from the IT world
- use web service technology as a powerful instrument for efficient BPM
- integrate business process models in various representational methodologies across organizations.

⁹ Because organizations normally use their own internal workflow languages that are independent of the workflow languages of other organizations, it is necessary to implement a uniform choreography interface to execute business process between collaborative organizations.

Table 1 Summary table of related works

Perspective	Research area & applying domain	Reference	Research objectives
Academic	Vision of SBPM	Hepp et al. 2005	To present a vision of SBPM to support agile and efficient BPM.
	SBPM architecture	Hepp and Roman 2007	To propose a representational architecture and ontology framework.
	SBPM lifecycle	Wetzstein et al. 2007	To identify new functionalities for SBPM based on the BPM lifecycle.
	Business process modeling	Benz et al. 2006	To develop an auto-completion mechanism to reduce typos and structural modeling errors using similarity measures between the manual business process and template business process.
		Lautenbacher and Bauer 2006	To develop an algorithm for automatic synthesis among modeled business processes.
		Thomas and Fellmann 2007	To remove the ambiguity caused by using natural language in semi-formal business process models by developing a semantic extension of the EPC.
	Business process implementation & execution	Cardoso and Sheth 2003	To suggest a method for efficient discovery of web services and automatic composition of those web services by developing a set of algorithms.
		Guo et al. 2004	To suggest a method that will support automatic discovery, composition, execution, and monitoring of web services using a conceptual mapping business process to semantic web service.
		Haller and Oren 2006	To develop m3pl ontology which increases the interoperability between collaborative organizations.
		Liao and Leung 2007	To reuse the information of existing business processes and web services and to accelerate the development of SOA application by developing a framework based on business process ontology (BPO).
Practical	Business process analysis	Celino et al. 2007	To introduce the technologies for semantic business process analysis and show how the technologies can benefit from the use of semantic information.
		Pedrinaci and Domingue 2007	To build ontologies for process monitoring and mining, Event Ontology (EO) and Process Mining Ontology (PMO).
	Logistics and supply chain	Preist et al. 2005	To develop a demonstrator system to apply semantic web service technology to logistics and supply chain.
		Drumm et al. 2006	To develop a message-mapping method to integrate the business process among parties in the logistic domain

4.2 Problems in the current practice of BPM

The tools and techniques used in the current practice of BPM are still incapable of satisfying stakeholders in the business world. Several problems that have been identified by the practitioners should be resolved to ensure successful implementation of BPM.

4.2.1 Difficulty in integrating business processes across departments and organizations

Integration of business processes, which has been identified as one of the most problematic issues associated with BPM, is becoming of greater concern as the number of business processes that should be managed intra-or inter-

organizationally increases. Business domain experts or process analysts design business processes in their own vocabularies and with different modeling methodologies. Although BPM tools provide formal methodologies and convenient environments required to design business processes, such inconsistent terms and heterogeneous modeling environments frequently cause conflicts, including name, notation and granularity conflicts, when business processes are integrated (Thomas and Fellmann 2007; Wetzstein et al. 2007). Therefore, **a uniform and standard way to specify business processes is required in order to integrate business processes across departments and organizations.**

4.2.2 Difficulty in querying and reusing all the facts associated with business processes

Current tools and techniques of BPM do not support capturing and representing all the semantic knowledge of business processes, and few process query languages¹⁰ allow both the business world and the IT world to traverse the BPS (Hepp et al. 2005). However, agile and efficient BPM requires the reuse of existing business processes defined in BPS, which can be satisfied if it is possible to query against BPS to find business processes relevant to carrying out business activities. Hepp et al. (2005) also suggested that sharing of BPS by the business and IT worlds should be possible without requiring manual efforts. Therefore, **there should be a unified BPS, shareable by both business and IT worlds**, so that domain experts and process analysts can query against all the information associated with business processes.

4.2.3 Non-automatic transformation from a business process model to an executable workflow model

Traditionally, a business process model in a high-level abstraction is good both for smooth communication among stakeholders in the business world and for expressive design of analytical process models. However, such an abstraction does not provide the technical information required for the implementation of business processes (Georgakopoulos and Hornick 1995; Stohr and Zhao 2001). In pursuit of efficient and agile process management, BPM aimed at reducing the time gap between the business world and the IT world by direct implementation and execution of business processes from the managerial perspective (Smith and Finger 2003). To that end, a number

of institutions (e.g., W3C, OASIS,¹¹ WfMC,¹² BPMI,¹³ and OMG¹⁴) and solution vendors (e.g., IDS-Scheer¹⁵ and E2E Technologies¹⁶) established specifications (e.g., XPDL,¹⁷ BPMN¹⁸ and BPEL) and developed tools (e.g. ARIS¹⁹ and Bridge4Process²⁰) for the automatic execution of business process models, by transforming them into executable workflow models, most of which are currently generated in BPEL. However, IT engineers must be involved in tuning the resultant BPEL documents, as BPEL cannot support all business process modeling notations (Mendling 2006, Verner 2004). This issue with current BPM solutions prevents the organization from responding to changes in the environment or organizational strategies in a timely fashion (Wetzstein et al. 2007). Therefore, **a more realistic and useful way to transform a business process model automatically into an executable workflow model is required.**

4.2.4 Irrelevant web services retrieved due to insufficient semantics of signature in WSDL specification

IT engineers frequently attempt to find web services that are relevant to the composition of a business process based on WSDL documents deployed on a Universal Description Discovery and Integration (UDDI) registry (Clement et al. 2004).

However, because WSDL specification allows only for syntactic descriptions of input and output parameters, a number of irrelevant web services may be retrieved (Cardoso and Sheth 2006; Studer et al. 2007). For example, there could be two web services with the same input and output parameters, but each may operate differently. As a result, WSDL documents must be interpreted manually by IT engineers. Therefore, **web services should be specified semantically, so that machines can correctly locate the relevant web services when they are needed.**

4.2.5 Lack of semantic description in BPEL specification for dynamic discovery and automatic composition of web services

BPM solutions can support automatic transformation from business process models to executable workflow models in

¹⁰ Business Process Query Language (BPQL) (Deutch and Milo 2007), a visual structural query language based on business process pattern, was developed in accordance with BPEL specification. However, BPQL does not support semantic queries because the BPEL specifies syntactic information to execute business processes.

¹¹ <http://www.oasis-open.org/home/index.php/> (Organization for the Advancement of Structured Information)

¹² <http://www.wfmc.org/> (Workflow Management Coalition)

¹³ <http://www.bpmi.org/> (Business Process Management Initiative)

¹⁴ <http://www.omg.org/> (Object Management Group)

¹⁵ <http://www.ids-scheer.com/>

¹⁶ <http://www.e2ebridge.com/>

¹⁷ <http://www.wfmc.org/XPDL/> (XML Process Definition Language)

¹⁸ <http://www.omg.org/docs/formal/08-01-17.pdf/> (Business Process Modeling Notation Specification)

¹⁹ http://www.ids-scheer.com/en/ARIS/ARIS_Solutions/5740.html/

²⁰ <http://www.e2ebridge.com/en/solutions/bridge-4-process.asp/>

BPEL. However, if a web service defined in a BPEL document is unavailable at run time, the business processes that use the web service come to a halt, which could be a potential hazard in terms of overall business performance. This sort of interruption can occur because BPEL specification allows web services in WSDL to be manually and statically assigned in design time,²¹ and represents only their syntactic information (Mendling 2006; Studer et al. 2007). The static dependency between BPEL specification and WSDL specification, along with the lack of semantics for its control flow, makes it impossible to discover appropriate web services and automatically compose them at run time. *Therefore, a semantic communication protocol for the dynamic discovery and automatic composition of web services is required to guarantee successful business process execution without process failure.*

Table 2 summarizes the major problems from which the current practice of BPM suffers, as well as the requirements for successful BPM.

5 Building SBPS for the sales order process

All the requirements derived in Section 4 can be met by defining SBPS as a collection of ontologies that provide the semantic information required for a successful implementation of SBPM. We first depict the overall structure of SBPS in Section 5.1 and then demonstrate how to build SBPS based on that structure with the example of a specific business domain, *Sales Order*.

5.1 The overall structure of SBPS

SBPS consists of two layers of semantic business process knowledge: *GBPO* and *SBPM/SWS*, as shown in Fig. 2. *GBPO* captures concepts associated with generic business process and the relationships among them. The concepts and relationships in *GBPO* are inherited to the business processes in *SBPM/SWS*. The *GBPO* consists of five sub-ontologies—*Business Process Ontology* (BPO), *Organization Ontology* (OO), *Resource Ontology* (RO), *Domain Knowledge Ontology* (DKO), and *Service Ontology* (SO)—each of which captures certain important concepts as follows:

- **BPO:** concepts for representing business processes that are performed in an organization or inter-organization
- **OO:** concepts associated with organizational units that are the performers of business processes

- **RO:** concepts associated with input and output resources of business processes
- **DKO:** concepts that represent specific domain knowledge helpful for performing business processes, such as payment condition, shipment condition, unit of measure, currency, and business rules
- **SO:** concepts that represent metadata of the semantic web service, including service signature and service location.

SBPM/SWS consists of three elements: *business process*, *semantic web service*, and *mapping between a business process and a corresponding semantic web service*.

5.2 Analysis of sales order process

From the several modules of the SAP R/3 ERP application,²² we analyzed the Sales and Distribution (SD) module and a part of the Materials Management (MM) module related to the sales order process. We examined the standard *activities* that are realized in the two modules prior to identifying a set of elementary *functionalities*, each of which can perform a business function. Table 3 provides some functionalities in terms of activities with input and output parameters, and Table 4 shows predefined *atomic* and *composite* business processes for a part of the sales order process based on the functionalities defined in Table 3. Each atomic business process has its corresponding functionality, which may have one or more activities that are always executed together. A composite business process is composed of two or more atomic business process and/or other composite business processes.

5.3 Building GBPO for sales order processes

5.3.1 Business process ontology (BPO)

BPO defines the concepts that are related to business processes themselves. Figure 3 shows the classes and their object properties defined in BPO. Ovals, dotted arrows and the symbol “ $\leftrightarrow$ ” represent classes, property, and inverse property, respectively. Classes in gray (e.g., *so:Service*, *oo:Organization* and *ro:Resource*) in the figure denotes that they are defined in external ontologies and the prefixes (e.g., *so*, *oo*, and *ro*) in front of the class names represent each namespace.

In Fig. 3, class *BusinessProcess* represents an abstract generic business process with two sub-classes: *AtomicBusiness-*

²¹ For example, prior to defining *partner link* in the BPEL specification to invoke a web service, one must define *port types* of the web service in the WSDL specification.

²² Since the SAP R/3 ERP application, which implements globally standardized and optimized business processes, includes comprehensive domain knowledge, we decided to analyze the business processes defined by the application.

Table 2 Summary of major problems in the current practice of BPM and requirements for successful BPM

Result of problem	Cause of problem	Requirement	References
Difficulty in integrating business processes	Business domain experts or process analysts design business processes using their own vocabularies, modeling methodologies and tools.	A uniform and standard way to specify business processes	Thomas and Fellmann 2007; Wetzstein et al. 2007
Difficulty in querying and reusing all the facts associated with business process	Current tools and techniques do not support the representation of various kinds of semantic knowledge about business process and few process query languages allow the business and IT worlds to traverse BPS.	A unified business process space that is shareable by both business and IT worlds	Hepp et al. 2005; Hepp and Roman 2007
Non-automatic transformation from a business process model to a workflow model	Most executable workflow models are currently generated in BPEL. However, IT engineers must be involved in tuning the resulting BPEL documents, since BPEL cannot support all business process modeling notations.	A more realistic and useful way to transform a business process model automatically into an executable workflow model	Mendling 2006; Verner 2004; Wetzstein et al. 2007
Irrelevant web services retrieved	WSDL specification allows only for syntactic descriptions of input and output parameters	Web services that are specified semantically.	Cardoso and Sheth 2006; Studer et al. 2007
Difficulty in dynamic discovery and automatic composition of web services	BPEL specification allows web services in WSDL to be assigned manually and statically in design time and represents only the syntactic information of a control flow of web services.	A semantic communication protocol for the dynamic discovery and automatic composition of web services	Mendling 2006; Studer et al. 2007;

Process and *CompositeBusinessProcess*. A business process takes some resources as input and produces new resources as output. An atomic business process is represented as a control flow of *business process elements*, each of which can be an *event*, *activity*, or *connector* in EPC.²³ An event is typically used to initiate a business process, to activate an activity, or to end a business process. An activity represents one or more tasks carried out by human agents or information systems to achieve a business goal and may produce an event. An event or an activity can be connected by one of three types of connectors: AND, XOR or OR. The connectors are used to connect events and activities logically by joining multiple incoming arcs as one outgoing arc or by splitting one incoming arc into multiple outgoing arcs in the control flow. In EPC, the typical form of an atomic business process is $(EL \rightarrow AL)^+ \rightarrow E$, in which EL is an event-list that may be one event or multiple events connected by connectors, AL is an activity list that can be one activity or multiple activities that may be preceded by connectors, X^+ indicates one or more repeated occurrences of X, and E is an event.

²³ Among a number of prominent formal methodologies including EPC and BPMN, This paper applies the EPC modeling methodology, which is used commonly by both the research community and industry.

Table 5 shows the typical elementary types of connections of events, activities, and connectors. More complex business processes can be represented by the combination of the connection types so, in order to design the business process correctly, these types of connections should be reflected in the BPO. For example, connection type 1 in Table 5 can be specified using the axiom of classes. The

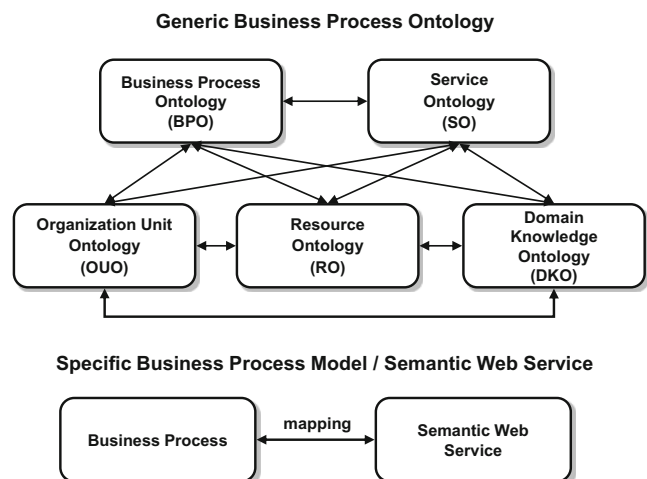
**Fig. 2** Semantic business process knowledge in SBPS

Table 3 Activities with input and output parameters for each functionality

Functionality	Activity	Input	Output
Checking available stock level of a sales order item on a required delivery date	Check ATP (Available-To-Promise) rule	Material ID, Sales order item, MRP element, MRP schedule	ATP rule for a sales order item
	Check the available stock level of a sales order item on a required delivery date	Material ID, Sales order item, Plant, ATP rule, MRP schedule, Unit of measure	Available stock quantity of a sales order item
Identifying the confirmed quantity	Calculate the confirmed quantity of a sales order item	Material ID, Sales order item quantity, Available stock quantity of sales order item, Unit of measure, Required delivery date	Confirmed quantity of a sales order item
Checking ATP achievement	Check ATP achievement	Sales order item number, Confirmed quantity of sales order item, Sales order item quantity	Result of checking ATP (i.e., achieved or not achieved)

For more information on all the partial descriptions and ontologies defined in this paper, refer to our research website, <http://ontology.korea.ac.kr/sbpm/ISF/index.html>

axiom of the class *Event* can be represented in *Description Logic* as follows:

$$\text{Event} \sqsubseteq \text{BusinessProcessElement} \sqcap \forall \text{activates.Activity} \\ \sqcap \exists \text{activates.Activity} \sqcap (= 1 \text{ activates})$$

This expression represents that, if something is a member of the class *Event*, 1) it should also be a member of the class *BusinessProcessElement*; 2) the instances of class *Event* have a relationship only to the instances in class *Activity*; 3) all the instances of class *Event* should have at least one specific relationship to the instances of class *Activity*; and 4) one instance of class *Event* has a relationship to exactly one instance of class *Activity*. By defining an axiom of class *Event* in this fashion, we can restrict the relationship between class *Event* and class *Activity* that is connected by the given property *activates*.

An instance of the class *BusinessProcess* has diverse relationships with instances of the classes *Service*, *Organization*, and *Resource*, each of which is connected by the properties *usesService*, *performs*, and *takesInput* and *producesOutput*, respectively. In particular, the property *usesService*, which relates the instances of the class *BusinessProcess* to those of the class *Service*, can be used to map business

processes and semantic web services, thereby making it possible to generate workflows automatically.

5.3.2 Organization ontology (OO)

OO consists of concepts that represent organizational units that perform business processes. In Fig. 4, class *Organization* has five sub-classes—*Company*, *Department*, *Plant*, *Storage*, and *SalesOrganization*—and has a recursive property *hasSubOrganization*, which indicates that an organization may have a sub-organization with its own sub-sub-organization, and so on.

Class *Company* represents the topmost organization, class *Department* a functional unit of a company, class *Plant* a factory or a warehouse, and class *Storage* the location where products and/or raw materials are stored. A company may have several departments, plants, and sales organizations as its sub-organizations. A plant may have a number of storages and MRP schedules (see Section 5.3.3). A sales organization should belong to a department, have a distribution channel (e.g., retail or wholesale) and a material division (see Section 5.3.3), and process sales orders placed by customers. Class *Position* represents the specific role performed in an organization, in an activity or in a business process by an individual represented as an

Table 4 Atomic and composite business process for a part of sales order process

Functionality	Atomic business process	Composite business process (Level 1)
Checking available stock level of a sales order item on a required delivery date	<i>CheckAvailableStockLevel-ABP</i>	<i>CheckATP-CBP</i>
Identifying the confirmed quantity	<i>IdentifyConfirmedQuantity-ABP</i>	
Checking ATP achievement	<i>CheckATPAchievement-ABP</i>	

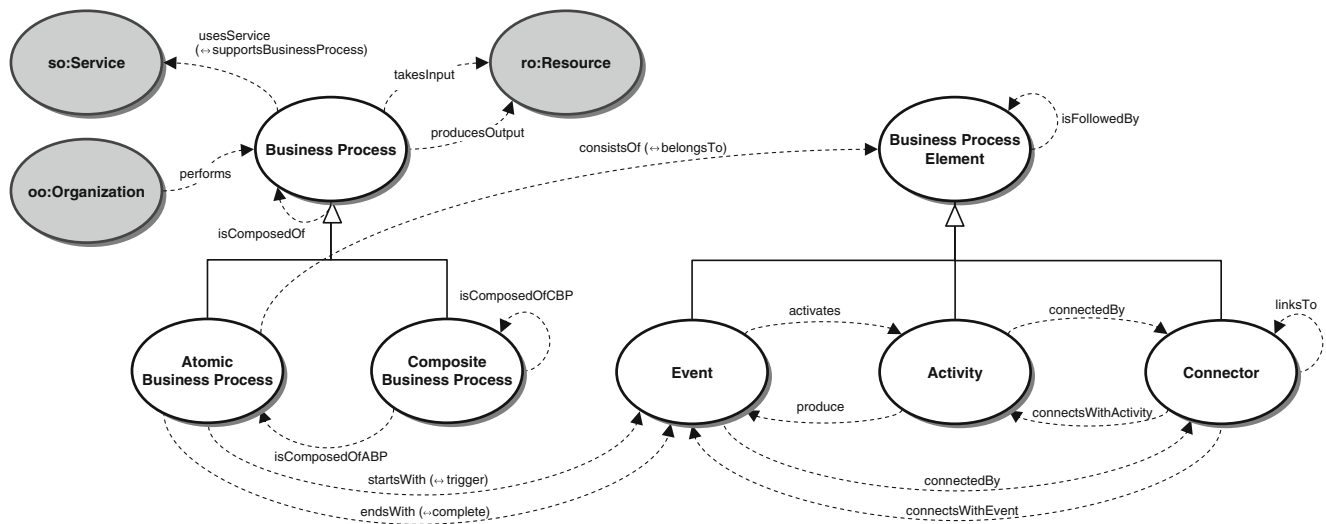


Fig. 3 Classes and object properties in BPO

instance of class *Personnel* in RO. Class *Location* is where a specific organization physically exists.

5.3.3 Resource ontology (RO)

RO consists of concepts that represent organizational resources²⁴ consumed by an organization to perform business processes. Figure 5 shows the classes and object properties defined in RO. Class *Resource* has three sub-classes: *Personnel*, *InformationSystem*, and *Material*. Class *Personnel* represents an individual who belongs to an organization but may occupy more than one position. Class *InformationSystem* which represents all the software applications and hardware systems used to store, retrieve, process and manage information within an organization, has five sub-classes: *D-Class*, *Service*, *ServiceInventory*, *Application*, and *InformationCarrier*. Among these sub-classes, the first three classes are associated with services and will be separately discussed in Section 5.3.5, where service ontology is described, as they perform essential and useful roles in the SBPM.

Class *Application* represents the software system used to carry out a business goal (e.g., sales order management system). Class *InformationCarrier* represents the system that stores and manages the information associated with business processes (e.g., a file system or database). An information carrier can be the platform for one or more business process applications.

Class *Material* represents raw material or goods. Class *MaterialDivision* represents a specific group of materials

from the sales point of view (e.g., a DVD player or MP3 player), while Class *MaterialType* represents the same type of materials from a manufacturing point of view (e.g., raw materials, finished goods, or trade goods). Class *MRPSchedule* represents a firm's production schedule based on customer order requests or demand forecasts. All materials should belong to a specific material division, have a material type and a unit of measure, and be stored in a specific storage. A material is usually produced according to an MRP schedule.

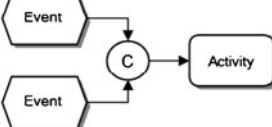
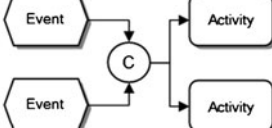
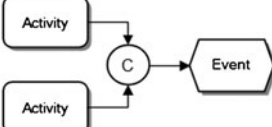
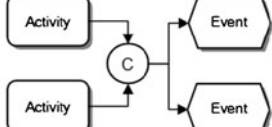
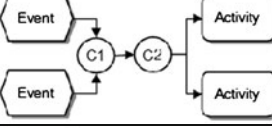
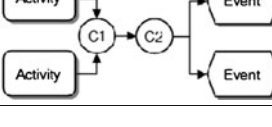
5.3.4 Domain knowledge ontology (DKO)

DKO consists of concepts associated with the business domain knowledge that is required for the sales order processes, including *payment condition*, *Incoterms*, *sales unit of measure*, *currency*, and *business rule*. Figure 6 illustrates a myriad of concepts associated with domain knowledge for the sales order process but, due to space limitation, we describe only major concepts in this section.

The two most important concepts included in the DKO are *SalesOrder* and *SalesOrder Item*. A single sales order should have one or more sales order items, and each item should belong to a single order. A sales order is requested by a customer represented as "sold-to-party" in the sales domain. A customer can often function as a different party, such as a party that represents where goods are to be sent (i.e., *ship-to-party*), one that represents where the bill should be sent for payment (i.e., *bill-to-party*), or one that represents who is paying the bill (i.e., *payer*). Class *Customer* defines these different functions of a customer by the property *hasCustomerFunction*. In some cases, each of the parties in a single order can be a completely different customer, so we have to indicate which customer corresponds to which party in an individual order. For that reason, we define the properties, *soldToParty*, *shipToParty*,

²⁴ Some of concepts in RO are defined by referring to the EPC—resource object types. Because more than 100 resource object types are defined in the EPC, we have selected the resource types that are the most suitable for our objective of managing business processes in the context of SOA.

Table 5 Elementary types of connection

No	Connection Type	Meaning
1		An event activates an activity.
2		An activity produces an event.
3		If one event or multiple events occur depending on a connector, an activity starts.
4		If one event or multiple events occur depending on a connector, multiple activities start.
5		If one activity or multiple activities are completed depending on a connector, an event happens.
6		If one activity or multiple activities are completed depending on a connector, multiple events happen.
7		If one event or multiple events occur depending on connector c1, one activity or multiple activities begin according to connector c2.
8		If one activity or multiple activities are completed depending on connector c1, one event or multiple events occur according to connector c2.

billToParty, and *payer*, as sub-properties of the property *hasCustomerFuction*.

Class *SalesOrderItem* represents each item of a single order. If necessary, each sales order item can be checked with a predefined *ATPcheckingRule* to verify whether the quantity of the sales order item will be available on the required delivery date. Class *ATPCheckingRule* is used when the ATP check is executed. Each sales order item is also arranged into one or more *plant* and *storage* for delivery.

Class *PaymentTerm* represents the payment conditions under which buyers pay for purchased goods (e.g., the period allowed for a buyer to pay off the amount due, and pay cash on delivery). Class *Incoterms* represents the shipping condition, such as FOB (Free On Board) and

CIF (Cost, Insurance and Freight). Class *IncotermsPort* represents the locations where the goods are shipped, and class *cur:Currency*²⁵ represents the type of money used in a particular country.

Domain knowledge normally includes a great deal of business rules. Such rules can be defined in Semantic Web Rule Language (SWRL) (Horrocks et al. 2004; Horrocks et al. 2005) which enables Horn clause-like rules to be added to semantic knowledge base in the form of a set of OWL axioms. Each rule in SWRL takes the form of *Antecedent*

²⁵ We reuse the *Currency* ontology (<http://www.daml.ecs.soton.ac.uk/ont/currency.owl>) developed by the University of Southampton in the UK.

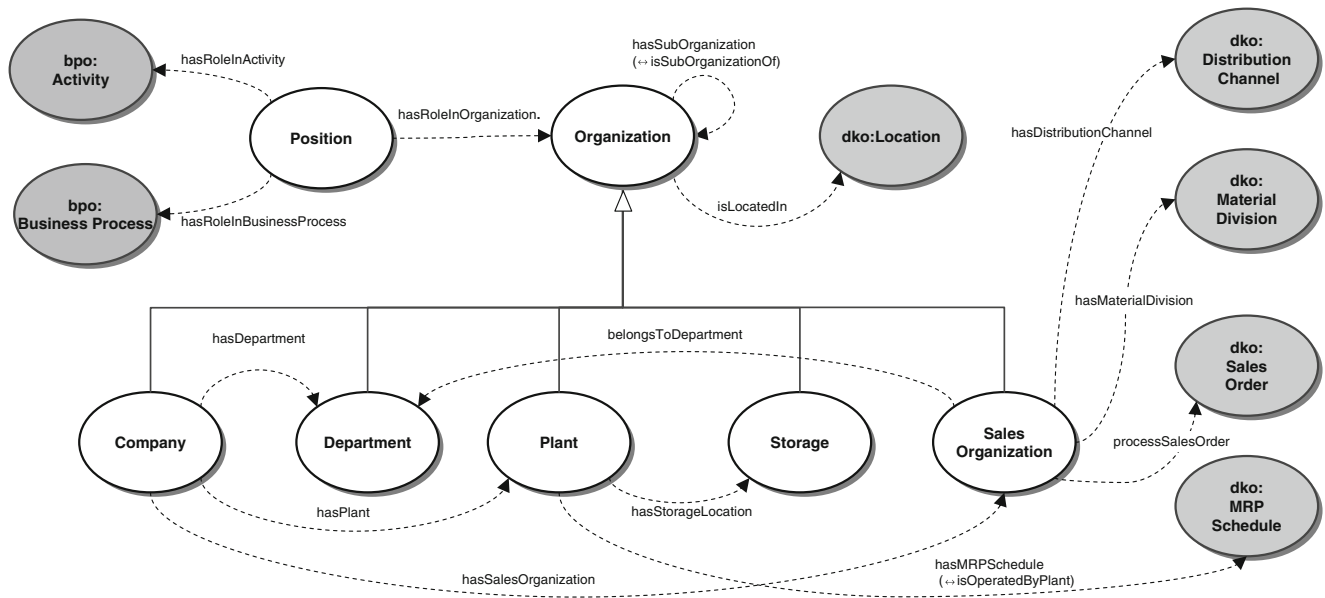


Fig. 4 Classes and object properties in OO

→ *Consequent*, where antecedent is a conjunction of atoms, and variables are letters prefixed by a question mark. An example rule is:

```
Customer(?a) ∧ hasCreditAmount(?a, ?b)
    ∧ swrlb : greaterThanOrEqual(?b, 1000000)
    → belongsToVIPCustomerGroup(?a)
```

This rule reads that if an amount of credit limit assigned to a customer is more than one million dollar, the customer belongs to a group of very important customers.

SWRL, however, has some limitations in representing various types of business rules. For example, SWRL does not (cannot) represent business rule as multi-arity predicates because it only supports representing it as unary or binary predicates. Besides, existing organizations have developed their own rule-based systems (e.g., expert systems) with different types of rule formats (e.g., production rule and reactive rule). For such reasons, the Rule Interchange Format (RIF) working group²⁶ started with developing a set of formats for interchange of rules in rule-based systems on the semantic web. RIF has a great deal of extensions to support features such as frames and objects, URIs, and XML schema data types. Although RIF is a part of semantic web architecture having compatibility with XML, RDF, OWL, and SPARQL, it is designed to enable interoperability among rule languages in general, thus not limited to the web. For these reasons, RIF seems to be appropriate for representing and interchanging a variety of business rules within or across organizations.

5.3.5 Service ontology (SO)

SO defines the concepts associated with the semantic web service (e.g., service signature, service inventory, business process performed by the service, and the applications on which the service runs). Normally, SO is used to integrate the management of services, reduce the redundancy of service development, increase the speed of service development, discover the relevant services, and generate workflows by mapping between business processes and corresponding services.

Figure 7 shows classes and object properties defined in SO. Class *Service* has four sub-classes: *DataService*, *BusinessService*, *VerificationService*, and *UtilityService*. Class *DataService* represents services associated with creation, retrieval, updating, and deletion of an organization's relevant data (e.g., customers, sales orders, invoices, and material data). Class *BusinessService* represents services associated directly with the logic for the performance of business processes (e.g., checking ATP achievement for a sales order item). Class *VerificationService* represents the service that verifies whether conditions requested from the *BusinessService* are true (e.g., checking available stock level). This type of service is often used in concert with *DataService* to complete a task requested from the *BusinessServices*. Class *UtilityService* represents services not directly associated with business process logic, including authorization, security, event-logging, and exception-handling. These types of services are highly beneficial when establishing a non-business-centric functional context.

Class *ServiceParameter* represents the parameters of an atomic service or a composite service. This class has two

²⁶ http://www.w3.org/2005/rules/wiki/RIF_Working_Group

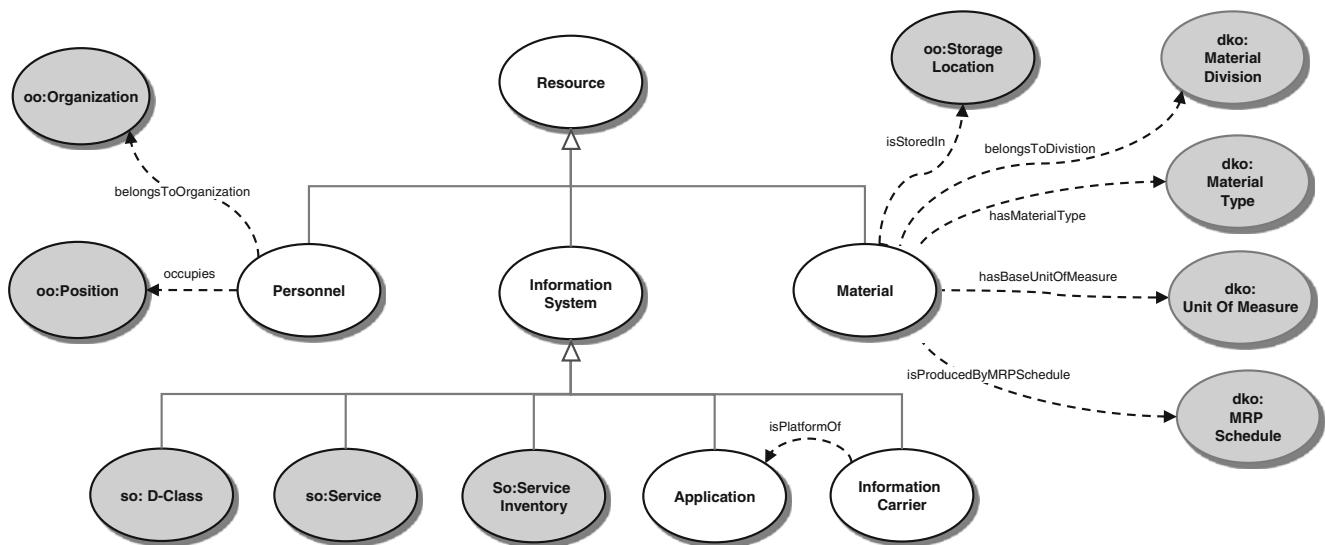


Fig. 5 Classes and object properties in RO

sub-classes, *ServiceInputParameter* and *ServiceOutputParameter*, which represent input and output parameters of a service, respectively. A service should have one or more input and output parameters. Properties *hasServiceInputParameter* and *hasServiceOutputParameter* represent the relationship between a service and the input parameters and the relationship between a service and the output parameters, respectively. Both input and output parameters should have a parameter type, which can be either a class

defined in the ontology or an XML Schema Definition (XSD) primitive data type, such as *xsd:string*, *xsd:integer*, or *xsd:double*.

Class *D-class* represents the metadata associated with all the classes defined in BPO, OO, RO and DKO, as well as their hierarchies. *D-class* can be used as the range of a parameter type when defining a semantic web service. If a class defined in the ontology is the range of an input or an output parameter type, then properties *hasD-classInputPar-*

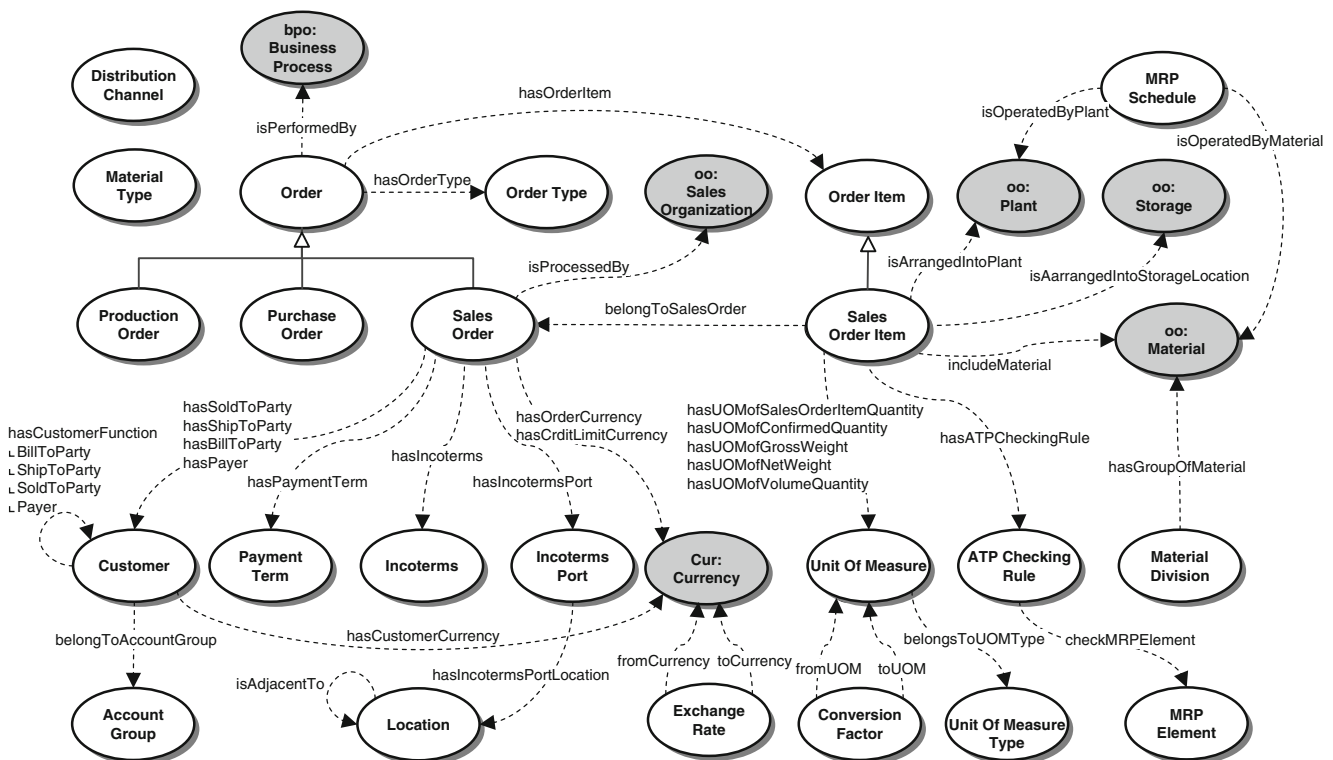


Fig. 6 Classes and object properties in DKO

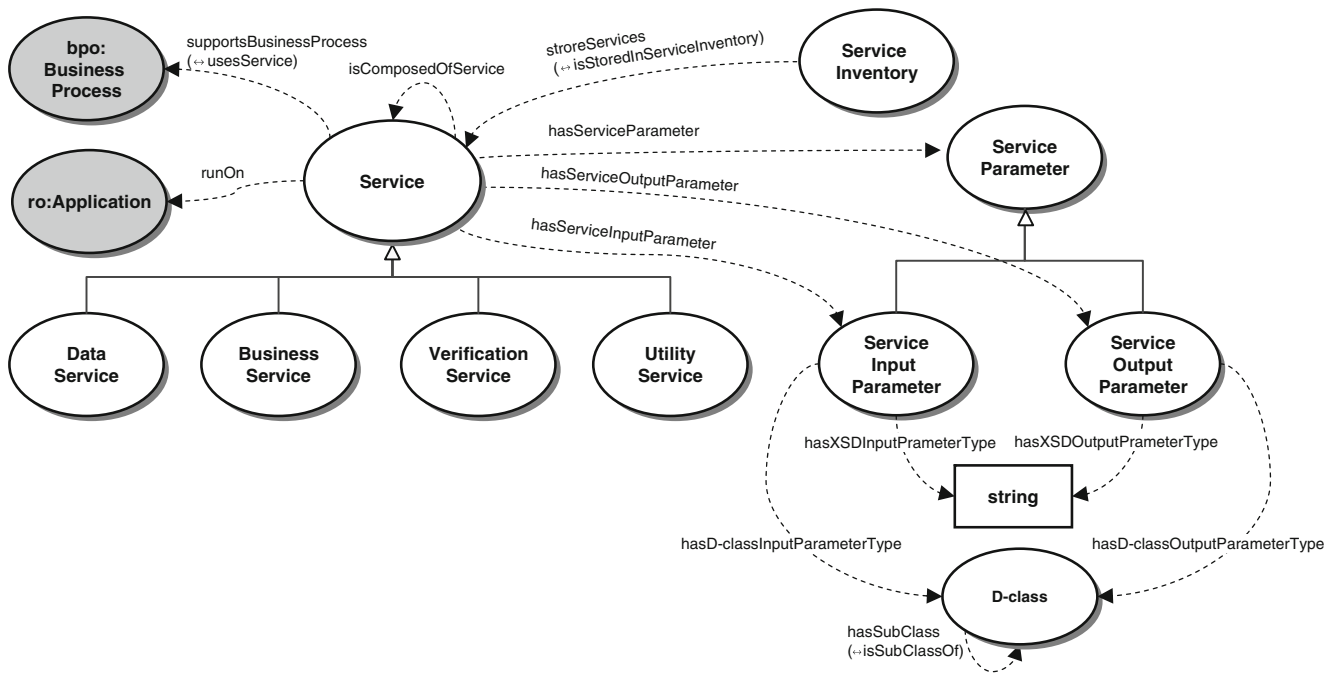


Fig. 7 Classes and object properties in SO

parameterType and *hasD-classOutputParameterType* relate an instance of class *ServiceInputParameter* and class *ServiceOutputParameter*, respectively, to a specific instance of class *D-class*. Class *ServiceInventory* represents a standardized and governed collection of services within or across an organizational boundary. Every service should be stored within the service inventory. Property *supportsBusinessProcess* is the inverse property of property *usesService* in the BPO and maps a service of class *Service* to a business process of class *BusinessProcess*.

5.4 Generating knowledge of SBPM/SWS

In this section, we explain how to generate knowledge of SBPM/SWS for the sales order processes using GBPO. SBPM/SWS consists of three types of knowledge: *business process*, *semantic web service*, and *mapping between the two*. Table 6 shows some mapping between business processes and their semantic web services.

Among the business processes in Table 6, we now demonstrate an example of generating the knowledge of

SBPM/SWS with the atomic business process *CheckAvailableStockLevel-ABP* which consists of three events and two activities as shown in Fig. 8. The business process calculates the stock level available on the required delivery date and, after checking a predefined ATP rule for a specific sales order item, returns the result as an output.

Based on GBPO, *CheckAvailableStockLevel-ABP* and *CheckAvailableStockLevel-ASWS* can be generated. A part of the definition of atomic business process *CheckAvailableStockLevel-ABP* is shown in Table 7, and that of atomic semantic web service *CheckAvailableStockLevel-ASWS* is shown in Table 8. Property *usesService* in Table 7 maps the business process *CheckAvailableStockLevel-ABP* to the semantic web service *CheckAvailableStockLevel-ASWS*. The other business processes and semantic web services should be defined similarly. In Tables 7 and 8, CLASS 1 is a super-class of CLASS 2, and INSTANCE is an instance of CLASS 2. Columns TYPE, NAME, NSD, DOMAIN, NSR, RANGE and VALUE indicate property type, property name, namespace of domain class, domain of property, namespace of range class, range of property and value of

Table 6 Mapping between business processes and their semantic web services

Atomic business process	Atomic SWS	Service type
<i>CheckAvailableStockLevel-ABP</i>	<i>CheckAvailableStockLevel-ASWS</i>	Verification Service
<i>IdentifyConfirmedQuantity-ABP</i>	<i>IdentifyConfirmedQuantity-ASWS</i>	Business Service
<i>CheckATPAchievement-ABP</i>	<i>CheckATPAchievement-ASWS</i>	Business Service
Composite Business Process (Level 1)	Composite SWS (Level 1)	
<i>CheckATP-CBP</i>	<i>CheckATP-CSWS</i>	

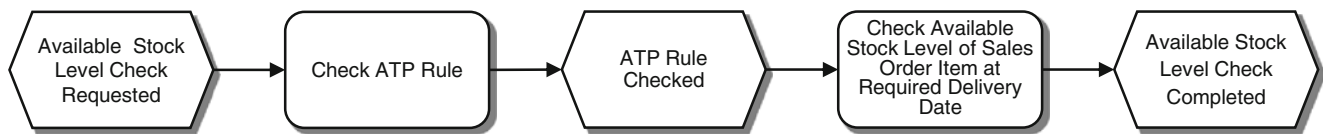


Fig. 8 An example of the business process: *CheckAvailableStockLevel-ABP*

property (literal value or instance), respectively. In column TYPE, **OP** and **DP** denote *ObjectProperty* and *Data-TypeProperty* in OWL, respectively. In column NSR, *bpo*, *ro*, and *so* refer to the prefixes of the ontologies, BPO, RO, and SO, respectively. Both the GBPO and SBPM/SWS are defined using Protégé version 3.3,²⁷ a Java-based ontology editor.

6 SBPS satisfies requirements for SBPM

This section provides a detailed explanation of how SBPS satisfies the requirements for SBPM that were derived from the problems in the current practice of BPM as discussed in Section 4.2.

6.1 A uniform and standard way to specify business process for business process integration

SBPS provides semantically schematic business process knowledge for resolving semantic and representational conflicts. A business process instance is normally created by composing business processes in SBPS but, if a predefined business process in SBPS is not available, a new atomic or composite business process should be defined and stored as business process in SBPS. Such knowledge of SBPS allows one to design uniform business process instances, thus facilitating the integration of business processes beyond organizational boundaries.

Figure 9 shows an example of creating a specific sales order process instance by using the SBPM/SWS knowledge in SBPS to compose three atomic business processes and five composite business processes.

6.2 A unified BPS, sharable by the business world and the IT world

SBPS also provides a basis for creating, querying, and reusing business process knowledge. All the business process knowledge registered in SBPS is represented in machine-understandable form,²⁸ such as RDF, thus allow-

ing stakeholders to derive benefits associated with business processes, including easy sharing and high interoperability due to uniform representation, as well as practical semantic query and manipulation.

Once SBPS is built, semantic queries from knowledge in SBPS are possible from both business (i.e., managerial) and technical perspectives. *Managerial* semantic queries include the following: *List all sales order processes performed by sales organization x*; *List all materials that are produced by plant x, included in material division y, and identified as material type z*; and *List all sales order processes that perform both an ATP check and a customer credit limit check*. Technical semantic queries include the following: *Show the control flow of services that perform sales order process y*; *Is there a service that can perform function x in service inventory y?*; and *List services in application y, related to process x*.

6.3 An automatic transformation from business process models into executable workflow models

SBPS contains mapping information between business processes and semantic web services. The ontology layer in Fig. 10 shows the ontologies, BPO, and SO, which are linked by the property *usesService* and its inverse property *supportsBusinessProcess*. These properties are used to generate mapping between business process and the corresponding semantic web service. The SBPM/SWS layer in Fig. 10 shows an example of mapping between a composite business process (e.g., *CheckATPAndSuggestAvailableDeliveryDate-CBP*) and a composite semantic web service (e.g., *CheckATPAndSuggestAvailableDeliveryDate-CSWS*). Each component and control flow of the business process is mapped to those of the semantic web services. The Workflow Instance level in Fig. 10 shows an example of a sales order workflow created using the composite semantic web service *CheckATPAndSuggestAvailableDeliveryDate-CSWS*.

6.4 Semantic web service and semantic communication protocol

SBPS provides a semantic description of a web service itself and communication protocol among a set of cooperating web services. We illustrate how the last two requirements in Section 4.2 can be satisfied with the implementation of the composite semantic web service *CheckATP-CSWS* using the

²⁷ <http://protege.stanford.edu/>

²⁸ All the knowledge on business processes in machine-understandable form is represented to stakeholders in the business world and the IT world as a form that is relevant to them using suitable tools for design, modification, execution, and analysis of the business process.

Table 7 A part of atomic business process *CheckAvailableStockLevel-ABP*

CLASS/INSTANCE			PROPERTY						
CLASS 1	CLASS 2	INSTANCE	TYPE	NAME	NSD	DOMAIN	NSR	RANGE	VALUE
Business_Process									
Atomic_Business_Process									
Check_Available_Stock_Level-ABP									
			DP	hasBusinessProcessID	bpo	Business_Process	xsd:string	ABP-003	
			DP	hasBusinessProcessName	bpo	Business_Process	xsd:string	Check_Available_Stock_Level-ABP	
			DP	hasBusinessProcessDescription	bpo	Business_Process	xsd:string	The business process calculates the stock level available on the required delivery date and, after checking a predefined ATP rule for a specific sales order item, returns the result as an output.	
					bpo	Business_Process	ro	D-class	DC_Sales_Order_Item
					bpo	Business_Process	ro	D-class	DC_Material
			OP	takeInput	bpo	Business_Process	ro	D-class	DC_MRP_Element
					bpo	Business_Process	ro	D-class	DC_MRP_Schedule
					bpo	Business_Process	ro	D-class	DC_Plant
			OP	producesOutput	bpo	Business_Process	ro	D-class	DC_Material
			OP	usesService	bpo	Business_Process	so	Service	Check_Available_Stock_Level-ASWS
					bpo	Atomic_Business_Process	bpo	Event	Available_Stock_Level_Check_Requested
					bpo	Atomic_Business_Process	bpo	Event	ATP_Rule_Checked
			OP	consistsOf	bpo	Atomic_Business_Process	bpo	Event	Available_Stock_Level_Check_Completed
					bpo	Atomic_Business_Process	bpo	Activity	Check_ATP_Rule
					bpo	Atomic_Business_Process	bpo	Activity	Check_Available_Stock_Level_of_Sales_Order_Item_at_Required_Delivery_Date
			OP	startsWith	bpo	Atomic_Business_Process	bpo	Event	Available_Stock_Level_Check_Requested
			OP	endsWith	bpo	Atomic_Business_Process	bpo	Event	Available_Stock_Level_Check_Completed

Table 8 A part of atomic semantic web service *CheckAvailableStockLevel-ASWS*

CLASS/INSTANCE			PROPERTY						
CLASS 1	CLASS 2	INSTANCE	TYPE	NAME	NSD	DOMAIN	NSR	RANGE	VALUE
Service									
Verification Service									
Check_Available_Stock_Level-ASWS									
			DP	hasServiceName	so	Service		xsd:string	Check_Available_Stock_Level-ASWS
			DP	hasServiceURI	so	Service		xsd:anyURI	http://ontology.korea.ac.kr/shpm/service/CheckAvailableStockLevel-ASWS.owl
			DP	hasServiceDescription	so	Service		xsd:string	This service first checks a predefined ATP rule for a specific sales order item. Then it calculates the stock level available on the required delivery date for a single sales order item and finally returns the result as an output
			OP	hasServiceInputParameter	so	Service	so	Service_Input_Parameter	In_Param_Available_Stock_Level_Check-ASWS
			OP	hasServiceOutputParameter	so	Service	so	Service_Output_Parameter	Out_Param_Available_Stock_Level_Check-ASWS
			OP	isStoredInServiceInventory	so	Service	so	Service_Inventory	Sales_Order_Service_Inventory
			OP	supportsBusinessProcess	so	Service	bpo	Business_Process	ABP-003
			OP	runsOn	so	Service	ro	Application	Sales_Order_Management_System

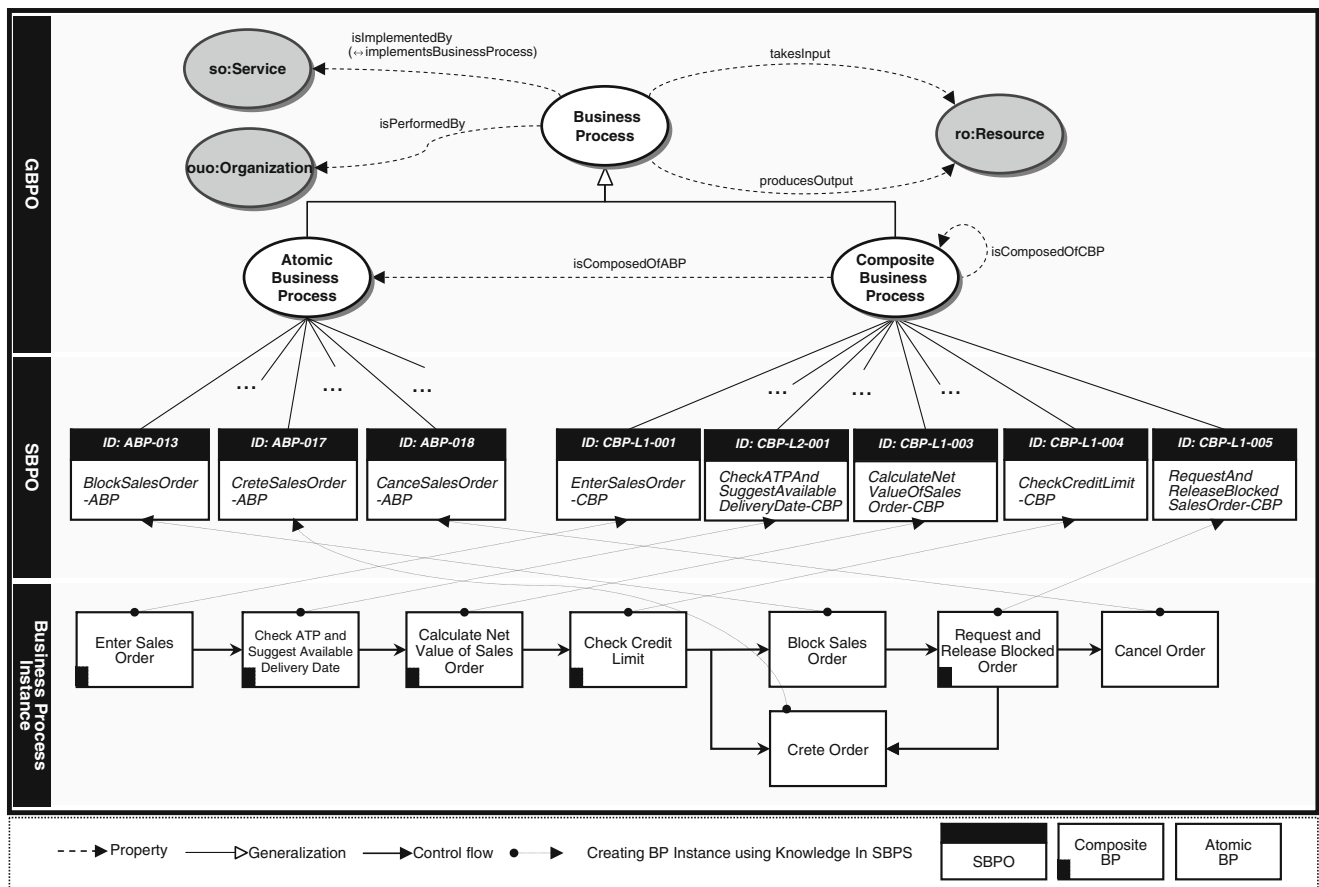


Fig. 9 A sales order business process created using SBPM/SWS knowledge in SBPS

knowledge in SBPS. We use *Protégé* Version 3.3, a prominent ontology editor, *OWL-S Editor*, a Tab Widget plug-in for *Protégé*, and *graphviz* Version 2.18, which visualizes the structural information of a semantic web service as diagrams of abstract graphs and networks.

CheckATP-CSWS is composed of three atomic services: *CheckAvailableStockLevel-ASWS*, *IdentifyConfirmedQuantity-ASWS* and *CheckATPAchievement-ASWS*. First, we define the input and output parameters of all atomic services. Table 9 shows a detailed signature of the *CheckATP-CSWS*, which consists of service type, service name, input parameter name, output parameter name, namespace of parameter, and parameter type. Types of input parameters or output parameters can be either XSD primitive data types or classes defined in ontology. By semantically specifying the service description using ontology, we enable the machine to understand what the input and output parameters mean and, as a result, can discover more relevant services that are deployed and advertised on the Web.

Figure 11 shows the flow of input and output parameters in the composite service *CheckATP-CSWS* in Table 9, which is expressed using *graphviz*. Normally, the output of a service can be taken as the input of a following service;

for example, the output (e.g., *Out_Available_Stock_Level*) of the atomic service *CheckAvailableStockLevel-ASWS* is taken as the input (e.g., *In_Available_Stock_Level*) of the following service *IdentifyConfirmedQuantity-ASWS*.

Figure 12 shows the screenshot after implementing the semantic web services *CheckATP-CSWS* using *OWL-S Editor*. *OWL-S Editor* facilitates the development of specifications for semantic communication protocol, as well as semantic description of the inputs, outputs, preconditions, and effects (IOPEs) of a web service. Figure 13 shows the control flow of the composite service *CheckATP-CSWS*, which was designed with *Visual Editor* in *OWL-S Editor*. As the control flow is specified in machine-understandable form, the machine understands the meaning of the communication protocol and can automatically compose and invoke relevant services while navigating a control flow.

7 Sales order process management in the environment of SBPS

This section illustrates a specific scenario to demonstrate how SBPM can be realized in an environment of SBPS.

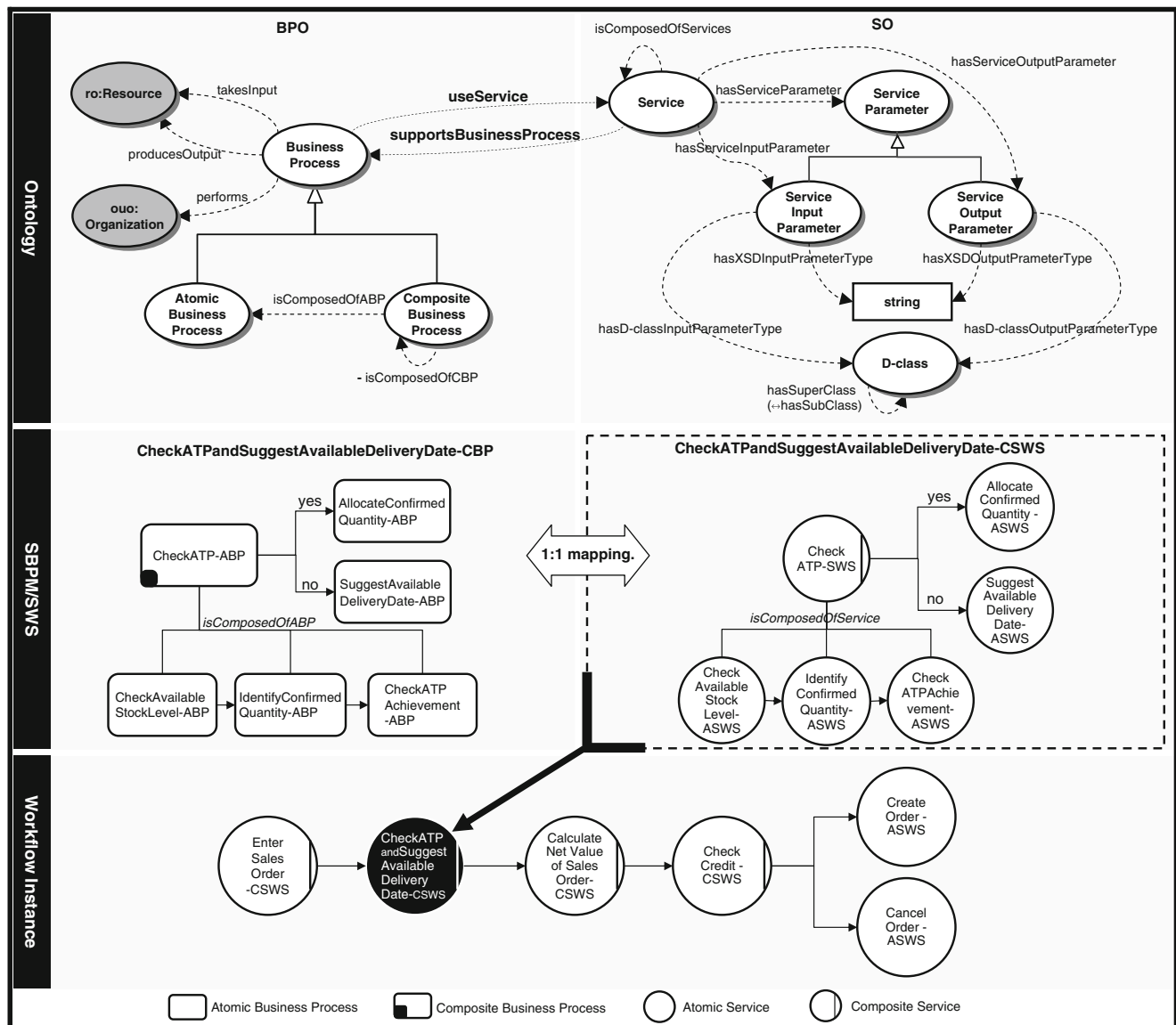


Fig. 10 An example of automatic transformation from business process into workflow

7.1 A scenario of managing a sales order process

Most companies face a scenario like this one when dealing with their sales order processes:

Company X produces and sells optical devices like CD-ROMs and DVD-ROMs. Recently, the company developed a new optical device model that supports higher resolution and more functionalities than its competitors' products. With this new product, the company expects to increase its dominance over the product market. Accordingly, Mr. Kim, Sales Director of the company, who is responsible for developing a new sales order process for the new product, searches the SBPS to locate reusable or modifiable existing business processes that include the following func-

tionalties; 1) **entering sales order information**, 2) **checking ATP and suggesting available delivery date**, and 3) **checking credit limit of a customer**.

Given this scenario, a list of tasks that covers the step, from design to execution, of a new business process for the sales order process for Company X can be written as follows:

1. Search SBPS for the existing business processes that can be used, with or without modification, for the execution of the three required functions (see Section 7.2.1).
2. If such business processes do not exist, define new business processes, a semantic web service corresponding to each business process and the mapping between the business process and the service, and then register the definitions into SBPS (as we did in Section 5.4).

Table 9 Signature of the service *CheckATPAndSuggestedAvailableDeliveryDate-CSWS*

Type	Service name	Input parameter name	Output parameter name	NS	Parameter type
A	<i>CheckAvailableStockLevel-ASWS</i>	-In_Material_ID		dko	Material
		-In_Sales_Order_Item		dko	Sales_Order_Item
		-In_MRP_Element		dko	MRP_Element
		-In_MRP_Schedule		dko	MRP_Schedule
		-In_Plant		oo	Plant
		-In_Unit_Of_Measure		dko	Unit_Of_Measure
			-Out_Available_Stock_Level	xsd	float
A	<i>IdentifyConfirmedQuantity-ASWS</i>	-In_Sales_Order_Item		dko	Sales_Order_Item
		-In_Available_Stock_Level		xsd	float
			-Out_Confirmed_Quantity	xsd	float
A	<i>CheckATPAchievement-ASWS</i>	-In_Sales_Order_Item		dko	Sales_Order_Item
		-In_Confirmed_Quantity		xsd	float
			-Out_ATP_Achievement_Result	xsd	boolean
C	<i>CheckATP-CSWS</i>	-In_Material_ID		dko	Material
		-In_Sales_Order_Item		dko	Sales_Order_Item
		-In_MRP_Element		dko	MRP_Element
		-In_MRP_Schedule		dko	MRP_Schedule
		-In_Plant		oo	Plant
		-In_Unit_Of_Measure		dko	Unit_Of_Measure
			-Out_Confirmed_Quantity	xsd	float
			-Out_ATP_Achievement_Result	xsd	boolean

3. Modify and/or compose the business processes (either found in step 1 or defined in step 2) as a new business process (see Section 7.2.2).
4. Generate a workflow to execute the new business process (see Section 7.2.2).

7.2 Realization of the scenario

7.2.1 Querying the existing business processes

In our scenario, Mr. Kim attempted to locate reusable and/or modifiable existing business processes by issuing a query in SPARQL (Prud'hommeaux and Seaborne 2007). As is shown in Fig. 14, Mr. Kim found several business processes that included most of the required functionalities. After examining each component of the retrieved business processes, he knew that the business process in the bold rectangle, *Sales_Order_Process_Retail_001*, was the most suitable to reuse when creating a new business process for the sales order process of the new product, because the other business processes included too many more functionalities than he needed, or too few. Subsequently, Mr. Kim verified that the first and the third functionalities mentioned in the scenario could be satisfied by *EnterSalesOrder-CBP* and *CheckCredit-CBP*, respectively, but the second could not. Therefore, he attempted to determine whether the

component business process *CheckATP-CBP* could be modified to satisfy the second functionality and learned that the *CheckATP-CBP* only checks whether the stock level for a sales order item will be available on the required delivery date.

However, the new business process requires that, if enough stock is available, the process should allocate the confirmed quantity of a sales order item and, otherwise, the process should suggest an available delivery date. Thus, Mr. Kim decided to modify the existing business process *CheckATP-CBP* into a new extended business process like *CheckATPAndSuggestAvailableDeliveryDate-CBP*, as described in step 3 of the task list.

7.2.2 Creating a new business process and generating a workflow

Mr. Kim identified the elementary functionalities based on the activities, and the atomic and composite business processes required to perform *CheckATPAndSuggestAvailableDeliveryDate-CBP* (see Table 10). Then, he defined mapping between business processes and semantic web services (see Table 11).

Based on the mapping, Mr. Kim added to the SBPS the knowledge of two atomic business processes, *AllocateConfirmedQuantity-ABP* and *SuggestAvailableDeliveryDate-*

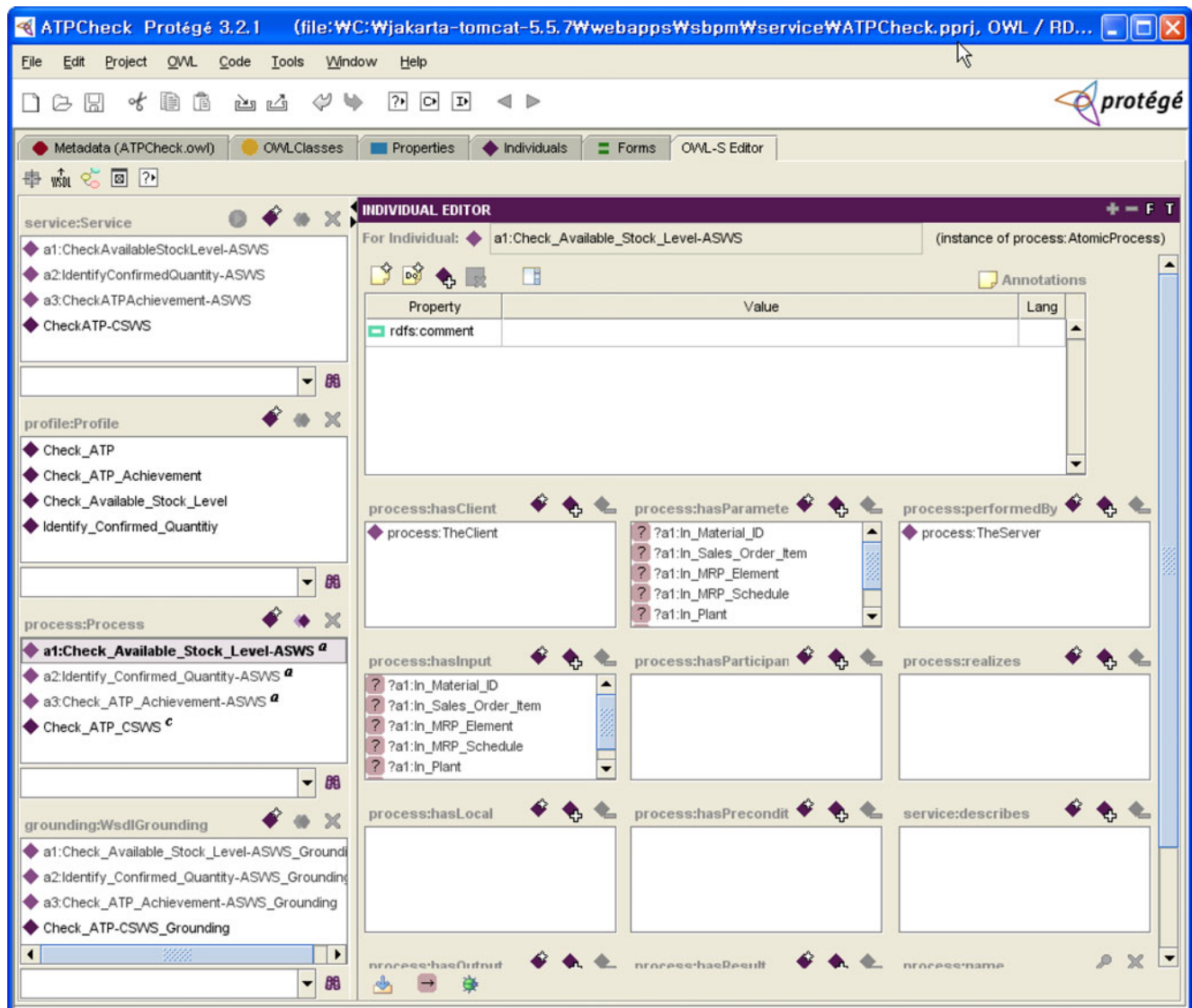


Fig. 12 A screenshot of OWL-S editor for defining *CheckATP-CSWS*

business processes, it is impossible for stakeholders in both the business world and the IT world to traverse the BPS in order to search for proper business process directly. In contrast, our SBPM approach demonstrated how to resolve the fundamental problems of the current practice of BPM

by adding sufficient semantics to BPS. We built SBPS by adding formal semantics to business processes as follows.

We first analyzed the sales order process, implemented as part of SD (Sales and Distribution) and MM (Materials Management) modules of SAP R/3 ERP Application. This pre-analysis allowed us to define the atomic and composite sales order processes, on the basis of which we implemented our business process ontologies, GBPO and SBPM/SWS.

We then demonstrated how the requirements for successful SBPM derived from the analysis of problems with current BPM practice can be satisfied by SBPS and explained how SBPM could be realized via a concrete scenario. In general, SBPS allows us to do things that would have been impossible otherwise, such as searching the business process space for all the facts associated with specific business processes necessary for the performance

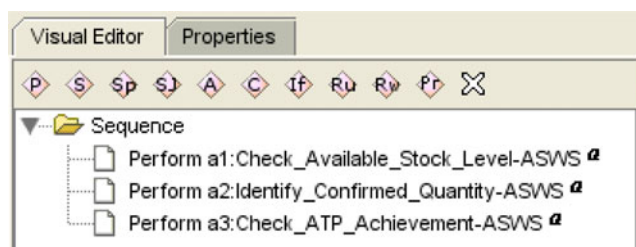
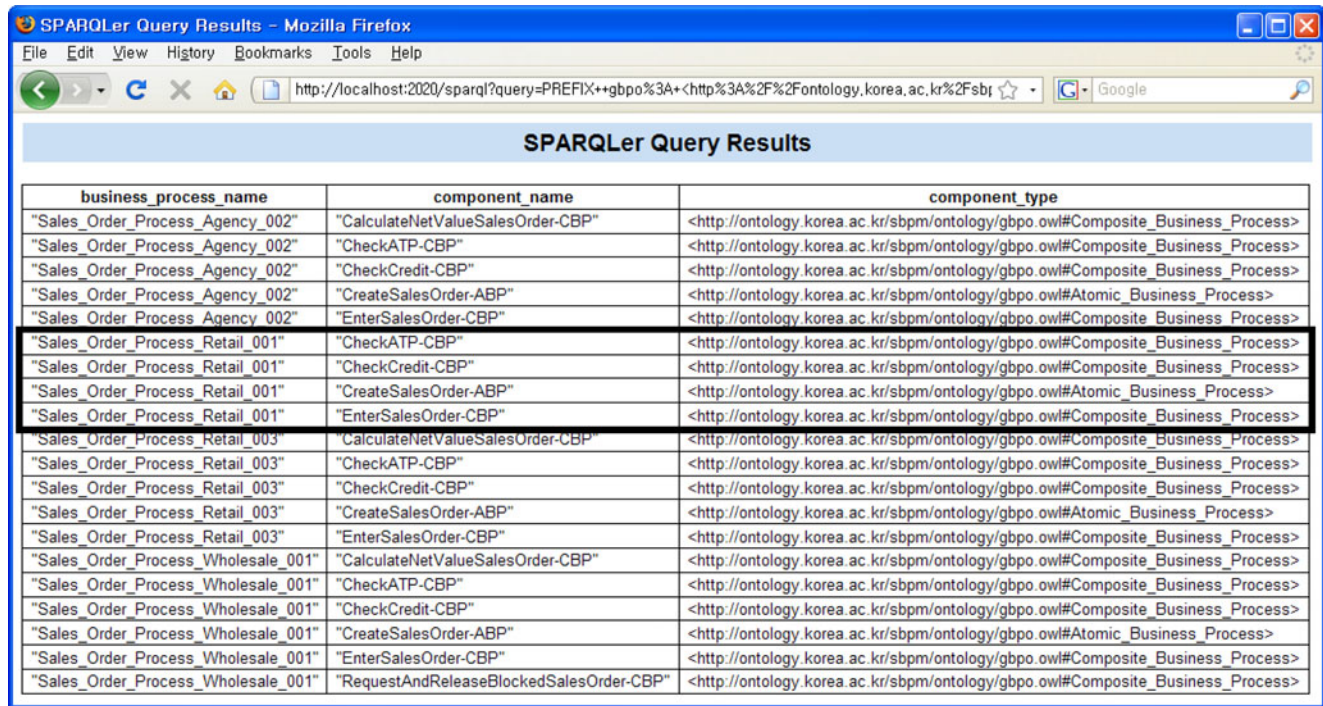


Fig. 13 A control flow of the service *CheckATP-CSWS*



business_process_name	component_name	component_type
"Sales_Order_Process_Agency_002"	"CalculateNetValueSalesOrder-CBP"	<http://ontology.korea.ac.kr/sbpm/ontology/gbpo.owl#Composite_Business_Process>
"Sales_Order_Process_Agency_002"	"CheckATP-CBP"	<http://ontology.korea.ac.kr/sbpm/ontology/gbpo.owl#Composite_Business_Process>
"Sales_Order_Process_Agency_002"	"CheckCredit-CBP"	<http://ontology.korea.ac.kr/sbpm/ontology/gbpo.owl#Composite_Business_Process>
"Sales_Order_Process_Agency_002"	"CreateSalesOrder-ABP"	<http://ontology.korea.ac.kr/sbpm/ontology/gbpo.owl#Atomic_Business_Process>
"Sales_Order_Process_Agency_002"	"EnterSalesOrder-CBP"	<http://ontology.korea.ac.kr/sbpm/ontology/gbpo.owl#Composite_Business_Process>
"Sales_Order_Process_Retail_001"	"CheckATP-CBP"	<http://ontology.korea.ac.kr/sbpm/ontology/gbpo.owl#Composite_Business_Process>
"Sales_Order_Process_Retail_001"	"CheckCredit-CBP"	<http://ontology.korea.ac.kr/sbpm/ontology/gbpo.owl#Composite_Business_Process>
"Sales_Order_Process_Retail_001"	"CreateSalesOrder-ABP"	<http://ontology.korea.ac.kr/sbpm/ontology/gbpo.owl#Atomic_Business_Process>
"Sales_Order_Process_Retail_001"	"EnterSalesOrder-CBP"	<http://ontology.korea.ac.kr/sbpm/ontology/gbpo.owl#Composite_Business_Process>
"Sales_Order_Process_Retail_003"	"CalculateNetValueSalesOrder-CBP"	<http://ontology.korea.ac.kr/sbpm/ontology/gbpo.owl#Composite_Business_Process>
"Sales_Order_Process_Retail_003"	"CheckATP-CBP"	<http://ontology.korea.ac.kr/sbpm/ontology/gbpo.owl#Composite_Business_Process>
"Sales_Order_Process_Retail_003"	"CheckCredit-CBP"	<http://ontology.korea.ac.kr/sbpm/ontology/gbpo.owl#Composite_Business_Process>
"Sales_Order_Process_Retail_003"	"CreateSalesOrder-ABP"	<http://ontology.korea.ac.kr/sbpm/ontology/gbpo.owl#Atomic_Business_Process>
"Sales_Order_Process_Retail_003"	"EnterSalesOrder-CBP"	<http://ontology.korea.ac.kr/sbpm/ontology/gbpo.owl#Composite_Business_Process>
"Sales_Order_Process_Wholesale_001"	"CalculateNetValueSalesOrder-CBP"	<http://ontology.korea.ac.kr/sbpm/ontology/gbpo.owl#Composite_Business_Process>
"Sales_Order_Process_Wholesale_001"	"CheckATP-CBP"	<http://ontology.korea.ac.kr/sbpm/ontology/gbpo.owl#Composite_Business_Process>
"Sales_Order_Process_Wholesale_001"	"CheckCredit-CBP"	<http://ontology.korea.ac.kr/sbpm/ontology/gbpo.owl#Composite_Business_Process>
"Sales_Order_Process_Wholesale_001"	"CreateSalesOrder-ABP"	<http://ontology.korea.ac.kr/sbpm/ontology/gbpo.owl#Atomic_Business_Process>
"Sales_Order_Process_Wholesale_001"	"EnterSalesOrder-CBP"	<http://ontology.korea.ac.kr/sbpm/ontology/gbpo.owl#Composite_Business_Process>
"Sales_Order_Process_Wholesale_001"	"RequestAndReleaseBlockedSalesOrder-CBP"	<http://ontology.korea.ac.kr/sbpm/ontology/gbpo.owl#Composite_Business_Process>

Fig. 14 Result of the SPARQL query example

of a specific task, reusing retrieved ones, composing them into a workflow model and executing that model without help from IT engineers.

The primary contribution of our research is the definition of ontologies associated with business process to build SBPS and the demonstration of how major problems discovered in the practice of current BPM can be resolved—or at least mitigated—using the ontologies in the environment of

SBPS. Our approach is novel and practical in that we attempted to add ontological annotation to globally standardized and optimized business processes captured from the analysis of SAP R/3 ERP application. Thus, our approach will give practitioners clear and useful insights into the successful implement of SBPM.

Our approach to analyzing business processes to define SBPM/SWS in SBPS may appear a costly, time-consuming

Table 10 Analysis of *CheckATP* and *SuggestAvailableDeliveryDate-CBP*

Activity	Functionality	Atomic business process	Composite business process (Level 1)	Composite business process (Level 2)
Check ATP (Available-To-Promise) rule	Checking available stock level of a sales order item on a required delivery date	<i>CheckAvailableStockLevel-ABP</i>	<i>CheckATP-CBP</i>	<i>CheckATP and SuggestAvailableDeliveryDate-CBP</i>
Calculate the confirmed quantity of a sales order item	Identifying the confirmed quantity	<i>IdentifyConfirmedQuantity-ABP</i>		
Check ATP achievement	Checking ATP achievement	<i>CheckATPAchievement-ABP</i>		
Allocate the confirmed quantity of a sales order item	Allocating the confirmed quantity of a sales order item	<i>AllocateConfirmedQuantity-ABP</i>		
Check MRP schedule for each sales order item	Suggesting available delivery date for each sales order item	<i>SuggestAvailableDeliveryDate-ABP</i>		

The gray areas in Tables 10 and 11 indicate newly added information that is required for the definition of the extended business process model *CheckATP and SuggestAvailableDeliveryDate-CBP* and the semantic web service *CheckATP and SuggestAvailableDeliveryDate-CSWS*

Table 11 Extended mapping for *CheckATPAndSuggestAvailableDeliveryDate-CBP*

Atomic business process	Atomic semantic web service	Service type
<i>CheckAvailableStockLevel-ABP</i>	<i>CheckAvailableStockLevel-ASWS</i>	Verification Service
<i>IdentifyConfirmedQuantity-ABP</i>	<i>IdentifyConfirmedQuantity-ASWS</i>	Business Service
<i>CheckATPAchievement-ABP</i>	<i>CheckATPAchievement-ASWS</i>	Business Service
<i>AllocateConfirmedQuantity-ABP</i>	<i>AllocateConfirmedQuantity-ASWS</i>	Business Service
<i>SuggestAvailableDeliveryDate-ABP</i>	<i>SuggestAvailableDeliveryDate-ASWS</i>	Business Service
Composite Business Process (Level 1)	Composite Semantic Web Service (Level 1)	
<i>CheckATP-CBP</i>	<i>CheckATP-CSWS</i>	
Composite Business Process (Level 2)	Composite Semantic Web Service (Level 2)	
<i>CheckATPAndSuggestAvailableDeliveryDate-CBP</i>	<i>CheckATPAndSuggestAvailableDeliveryDate-CSWS</i>	

and complex task. However, it should be noted that while the analysis of business processes in an organization is inherently quite complex, the resultant knowledge in SBPS, once created and continually managed, can be extremely

useful to stakeholders both in the business world and the IT world. Our approach allows us to achieve agile and efficient BPM by reducing the time and cost involved in developing a new business process in a dynamic business

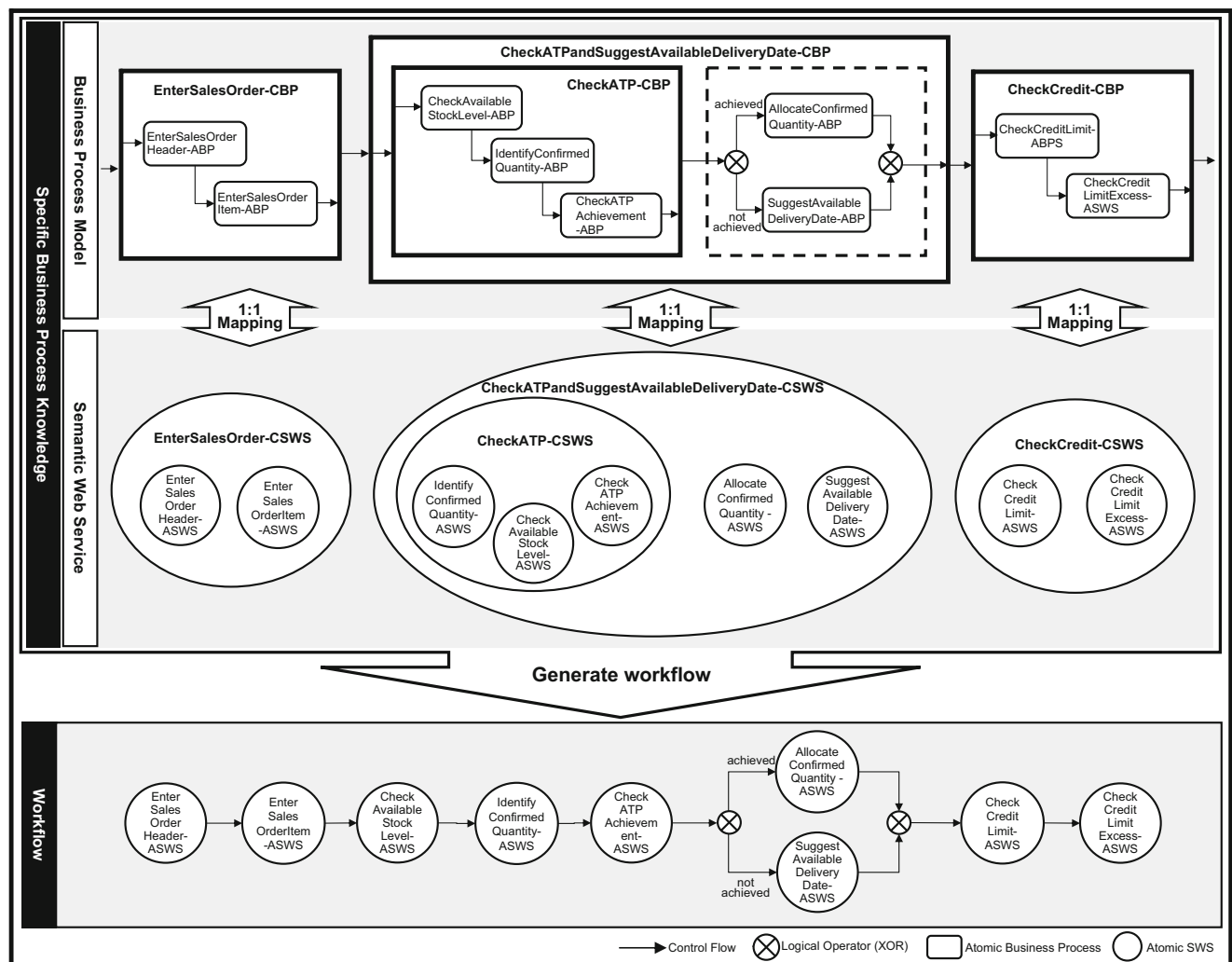


Fig. 15 Generation of workflow for the sales order process. At the business process model level, the area in solid lines indicates reused existing business models and the area in dotted lines indicates the new business model added to SBPS

environment. Additionally, we have used a specific modeling notation, EPC methodology, chosen among several modeling notations such as BPMN and UML, as the most expressive notation or language, so that we can represent any business process in that notation or language, however complex the business process may be. Although our paper defined semantic business process knowledge for a limited domain, our approach can be readily extended to other areas of business without posing a scalability problem.

This study leaves something to be desired in terms of a complete implementation of SBPM. Although SBPS defined in our research provides semantically richer information on business processes, we did not show how we can search for business processes and web services through semantic matching using this semantically richer information.

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